

1. Administrative & Study Identification

Full Study Title: A Clinical Study of Ifinatumab Deruxtecan Based Treatment Combinations or as Monotherapy to Treat Metastatic Castrate Resistant Prostate Cancer (mCRPC) (MK-2400-01A/IDEATE-Prostate02) **Short Title/Acronym:** IDEATE-Prostate02 / MK-2400-01A **Master Protocol:** MK-2400-U01 **Protocol Number:** MK-2400-01A **NCT Number:** NCT06863272 **Posting ID:** 791777209618820765 **Trial Status:** Recruiting **Sponsor/Company Name:** Merck **Investigational Product:** Ifinatumab deruxtecan (I-DXd, MK-2400), an investigational B7-H3-directed antibody-drug conjugate (ADC) **Phase of Development:** Phase I/II **Medical Monitor:** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: The purpose of this substudy is to assess the efficacy and safety of ifinatumab deruxtecan (I-DXd), given alone or with other treatments in participants with metastatic castration-resistant prostate cancer (mCRPC). The specific goals are to evaluate the safety of the study treatment and tolerability, determine a safe dose level of I-DXd that can be used with other treatments, and monitor participant levels of prostate specific antigen (PSA) during treatment. **Secondary Objectives:** To evaluate the objective response rate (ORR) and survival evaluations, including overall survival and radiographic progression-free survival (rPFS).

2.2 Background & Rationale There is a high unmet clinical need for effective therapies to improve outcomes from mCRPC once the cancer no longer responds to standard hormone-lowering treatments. I-DXd is a targeted antibody-drug conjugate that attaches to the B7-H3 protein found on many prostate cancer cells to deliver a potent topoisomerase I inhibitor payload directly to the tumor. Previous studies demonstrated promising antitumor activity, and this umbrella study aims to identify the optimal dose and combination treatments to maximize patient benefit.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (e.g., Docetaxel arm) or Single-arm, depending on the specific substudy phase **Blinding:** Open-label **Randomization:** Stratified randomization across the multi-arm umbrella substudy

3.2 Planned Enrollment & Duration Target \$\$\$: 360 participants
Number of Sites: Multicenter (Global) **Estimated Study Start:** 03-Jul-2025 **Estimated Primary Completion:** 18-Feb-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult males with a histologically- or cytologically-confirmed adenocarcinoma of the prostate without small cell histology. 2. Prostate cancer progression while on androgen deprivation therapy (ADT) (or post bilateral orchiectomy) within 6 months before Screening. 3. Current evidence of metastatic disease. 4. Has received prior treatment with 1 or 2 androgen receptor pathway inhibitors (ARPIs) and progressed during or after treatment. 5. Participants receiving bone resorptive therapy (including, but not limited to bisphosphonate or denosumab) must have been on stable doses for ≥ 4 weeks before allocation/randomization. 6. An Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 assessed within 10 days before allocation/randomization. 7. Has prior treatment with poly-ADP-ribose polymerase inhibitors (PARPi) if indicated by local approved regimen or were deemed ineligible to receive PARPi by the investigator.

4.2 Exclusion Criteria 1. History of (non-infectious) interstitial lung disease (ILD)/pneumonitis that required steroids, or has current ILD/pneumonitis or suspected ILD/pneumonitis. 2. Clinically severe pulmonary compromise resulting from intercurrent pulmonary illnesses. 3. Uncontrolled or significant cardiovascular disease. 4. History of pituitary dysfunction. 5. Poorly controlled diabetes mellitus. 6. History or current condition of adrenal insufficiency (e.g., Addison's disease). 7. Has received prior treatment with a taxane-based chemotherapy agent (e.g., docetaxel, Docefrez, Taxotere) for metastatic castration-resistant prostate cancer (mCRPC). 8. Chronic steroid treatment (dose of >10 mg daily prednisone equivalent), except for low-dose inhaled steroids (for asthma/COPD), topical steroids (for mild skin conditions), or intra-articular steroid injections. 9. Received prior radiotherapy within 2 weeks of start of study intervention, or has radiation-related toxicities, requiring corticosteroids. 10. Known additional malignancy that is progressing or has required active treatment within the past 3 years. 11. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 12. Active autoimmune disease that has required systemic treatment in the past 2 years. 13. History of allogeneic tissue/solid organ transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered primarily every 3 weeks (q3w) for monotherapy or according to

specific combination cohort schedules. **Route of Administration:** Intravenous (IV) for I-DXd; oral (pills) for combination ARPIs. **Duration of Treatment:** Continued until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Bone-strengthening drugs, provided the participant is on a stable dose for at least 4 weeks prior to screening. **Prohibited:** Chronic systemic steroid treatment (dose > 10 mg daily prednisone equivalent) and prior allogeneic tissue/solid organ transplants.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor Sample, CT/MRI/Bone Scans, Physical Exams, Labs (including baseline PSA).
Baseline	Day 0	Randomization, First Dose (Safety Lead-in or Efficacy Phase).
Interim	Every 3 Weeks	Safety Labs, Efficacy Assessment, PSA tracking, side-effect checks.
End of Study	Variable	Final Vital Signs, Exit Interview, End of Treatment Imaging.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- Efficacy Phase: Number of Participants Who Experience One or More Dose-Limiting Toxicities (DLTs) - Combination Arms Only.** DLTs include specified Grade 3/4 nonhematologic or hematologic toxicities, prolonged treatment delays, missing >25% doses in Cycle 1, or Grade 5 toxicities.
- Efficacy Phase: Number of Participants Who Experienced an Adverse Event (AE).**
- Efficacy Phase: Number of Participants Who Discontinued Study Intervention Due to an AE.**

7.2 Secondary Endpoints

- Objective Response Rate (ORR).
- Radiographic Progression-Free Survival (rPFS).
- Overall Survival (OS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via physical exams, blood/urine tests, with a specific focus on identifying Interstitial Lung Disease (ILD) and pneumonitis. **Serious Adverse Events (SAEs):** Routine 24-hour reporting timeline for severe toxicities. **Data Monitoring Committee (DMC):** Internal/External safety review of dose-limiting toxicities during the Safety Lead-in phase before expanding to the Efficacy Phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for efficacy endpoints; Safety Set for all adverse event tracking.

Sample Size Justification: Safety lead-in cohorts powered to identify Phase 2 dose recommendations; Efficacy cohorts powered to detect statistically significant improvements in response rates and survival. Total targeted enrollment is 360 participants.

Handling of Missing Data: Standard right-censoring for survival data and Multiple Imputation for missing continuous lab variables per oncology guidelines.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits for data verification and safety reviews. **Audit Trail:** Secure electronic Data Capture (eDC) system featuring eCRF locking and audit tracking.

11. Study Identifiers & Governance

Primary ID: MK-2400-01A **Posting ID:** 791777209618820765 **Registry Number:** NCT06863272 **Sponsor/Company:** Merck **Study Title:** A Clinical Study of Ifinatamab Deruxtecan Based Treatment Combinations or as Monotherapy to Treat Metastatic Castrate Resistant Prostate Cancer (mCRPC) (MK-2400-01A/IDEate-Prostate02) **Official Regulatory Title:** MK-2400-01A Substudy: A Phase 1/2, Open-label Umbrella Substudy of MK-2400-U01 Master Protocol to Evaluate the Safety and Efficacy of Ifinatamab Deruxtecan-based Treatment Combinations or Ifinatamab Deruxtecan Alone in Participants With Metastatic Castration-resistant Prostate Cancer (mCRPC) (IDEate-Prostate02)

12. Clinical Framework (Trial Specific)

Disease Areas: Castration-Resistant Prostatic Cancer, Metastasis (Preferred UMLS: Prostatic Neoplasms, Castration-Resistant; Neoplasm Metastasis)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** Confirmed adenocarcinoma, progression on 1-2 prior ARPIs, and active metastases

Medical Methodology: Antibody-Drug Conjugate (ADC) targeted therapy combinations

Optimization Target: Identification of I-DXd optimal dose and tracking of safety/DLTs and PSA reductions

13. Study Procedures & Methodology

Management Pathway: Standard oncology clinical pathway for mCRPC
Education Protocol (HCP): Investigational product administration and ILD/pneumonitis AE monitoring training
Education Protocol (Patient): Onsite training on symptom reporting (especially respiratory symptoms) and treatment schedule compliance
Quality Audit Mechanism: Tumor response imaging review (RECIST v1.1/PCWG3) and dose-limiting toxicity monitoring

14. Observation & Follow-Up Schedule

Total Duration: Up to primary completion in February 2030 (approx. 240 weeks)
Onsite Visit Cadence: Every 3 weeks (q3w) for treatment cycles
Remote Monitoring Frequency: Continuous safety monitoring between cycles
Checklist Review Interval: Tumor imaging (CT/MRI/bone scans) every 6 to 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				